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Recent advances in glycoinformatic platforms for glycomics and glycoproteomics

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Protein glycosylation is the most complex and prevalent post-translation modification in terms of the number of proteins modified and the diversity generated. To understand the functional roles of glycoproteins it is important to gain an insight into the repertoire of oligosaccharides present. The comparison and relative quantitation of glycoforms combined with site-specific identification and occupancy are necessary steps in this direction. Computational platforms have continued to mature assisting researchers with the interpretation of such glycomics and glycoproteomics data sets, but frequently support dedicated workflows and users rely on the manual interpretation of data to gain insights into the glycoproteome. The growth of site-specific knowledge has also led to the implementation of machine-learning algorithms to predict glycosylation which is now being integrated into glycoproteomics pipelines. This short review describes commercial and open-access databases and software with an emphasis on those that are actively maintained and designed to support current analytical workflows.

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Introduction

Protein glycosylation is involved in the vast majority of cellular interactions and complex networks and is the basis of most biological events, including protein stability and recognition [1]. It is commonly stated that more than 50% of all proteins are modified by glycans, although glycosylation is an untemplated process, it has been estimated that fewer than 3000 N-linked and O-linked exist in humans [2–5]. Changes in protein glycosylation play a major role in both health and disease including cancer metastasis, Alzheimer's, inflammatory responses, and host–pathogenic recognition events [6,7,8••,9,10,11••,12], with the abundances of different glycoforms varying over time, from individual-to-individual and within different cell environments [13,14]. Despite the connection between their structure and function, the glycoproteome remains understudied because of the analytical challenges associated with glycoproteomics (analysis of intact glycopeptides), and the inherent complexity of oligosaccharides [15]. These are key roadblocks that need to be overcome for glycosciences to advance and require fast, precise and robust analytical technologies and workflows that are data-rich with supporting glycoinformatics platforms [16–19].

Often combinations of methodologies and analytical techniques are used to elucidate structural information of glycans and their conjugates. These include liquid chromatography (LC), capillary electrophoresis (CE), nuclear magnetic resonance (NMR), mass spectrometry (MS), and (micro)arrays. Such technologies are becoming more standardised and streamlined helping to initiate efforts to capture the molecular complexity displayed by the glycoproteome. More recently advances have been made in developing analytical techniques that have the potential to address this complexity based on NMR and MS technologies including ion mobility spectrometry and IR spectroscopy [20,21,22,23•,24,25,26,27•,28,29]. The use of such technologies will not be discussed, and readers are referred to recent reviews [30•,31,32,33].

Data processing, annotation, analysis, and storage can often lead to major hurdles in large-scale studies [34•,35], and given the impact of data sets being generated efforts in glycoinformatics have steadily increased over the last few years. In this short review we discuss recent advances in glycoinformatics platforms that i) support the qualitative

and quantitative profiling of glycan structures released from glycoproteins and attached to fragmented glycopeptides using LC, CE and MS technologies, ii) databases that store structure and experimental data, and iii) machine-learning algorithms for glycosylation site prediction. Finally, we highlight developments in data reporting guidelines that are improving our ability to process and interpret data collections (Figure 1).

Software applications for the interpretation of profiling data

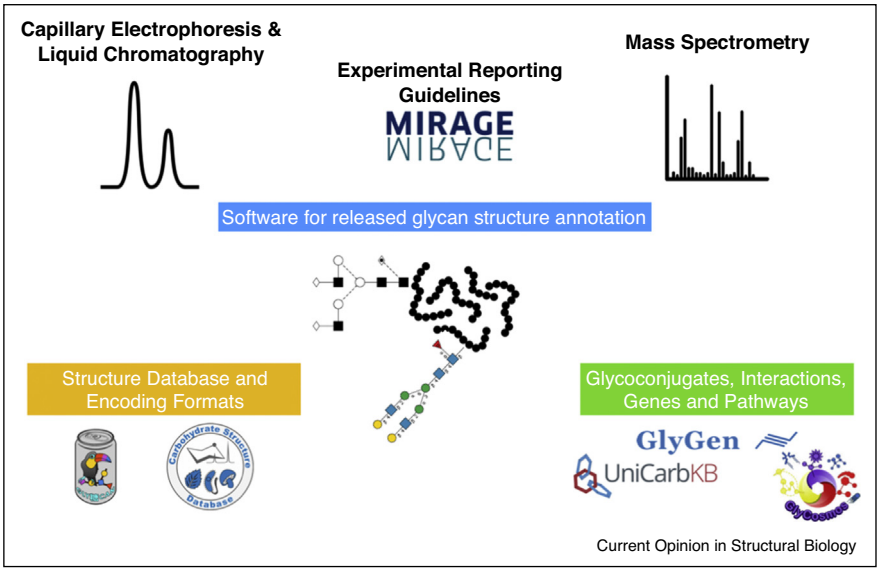
Quantitative glycoprofiling can aid in our understanding of the biological roles of glycans in disease progression. As described by Hua *et al.*, a unique feature of glycomics data is the heterogeneity of protein glycosylation, which provides an opportunity to identify many features that can be used to classify samples as diseased or healthy [36•]. Previously, we and others reviewed software devoted to released glycan MS analysis [37,38] and in the following sections, we describe new applications available for interpreting LC profiles and CE electropherograms (Table 1).

Liquid chromatography

Ultra-High Performance Liquid Chromatography (U/HPLC) has been at the forefront of analytical techniques used to separate, isolate and characterise oligosaccharides. In general, this involves converting reducing-end tagged glycans retention times to glucose units (GU) whereby the elution positions of glycans is determined by comparison with the elution positions of a standard mixture of glucose oligomers (commonly referred to as a dextran ladder), which is run alongside sample runs [39,40]. Typically, a fifth-order polynomial distribution curve is fitted to the dextran ladder to determine GU values from retention times, that can be compared to reference libraries or databases, for example, GlycoStore [41,42]. This approach has been widely used to minimise separation variability often observed across different instruments and columns [43] — a number of workflows (and comparison of methods) have been published for the analysis of serum protein glycosylation, single proteins and samples of varying complexity [44,45,46,47••,48,49,50,51].

In the last few years, improved sample preparation workflows and labelling reagents have decreased the time required for *N*-glycan sample preparation and enhanced the sensitivity of glycan detections, for example, RapiFluor-MS (RFMS) [52] and procainamide [53]. Importantly, Akmačić *et al.*, described the development and optimisation of high-throughput sample preparation methods noting that the clean-up of labelled glycans was the largest source of analysis variation [54]. In addition, the repertoire of exoglycosidases routinely used from structural characterisation has expanded with a more specific fucosidase and endoglycoceramidase [55•,56]. Improvements to sample preparation and the development of high-throughput platforms as led to increased demand for tools that overcome the time-consuming task of data analysis [57,58•,59].

Figure 1



The purpose of this short review is to describe advances in software applications, databases and reporting guidelines with a focus on released glycans. The figure shows the five topics described in this review including: 1) resources for interpreting LC and CE data; 2) structural databases and frameworks for handing computational encodings; 3) international activities focused on glycoconjugates and the integration of glycan-related data from 'omic databases; and 4) reporting guidelines. The figure shows a limited selection of resources, refer to Tables 1–3 for more descriptions.

Table 1

Software applications for analysing LC and CE data collections

Name	Description	Link	Reference
Liquid Chromatography			
GlycoProfileAssigner	Assigns structures to UPLC profiles by matching glycan structures with GU values and sequential exoglycosidase digestions.	https://bitbucket.org/fergaljd/glycoprofileassigner	[60]
GlycanAnalzyer	Interpretation of *N*-glycan UPLC profiles with exoglycosidase digestions.	http://glycananalyzer.neb.com	[65]
HappyTools	Standalone application for high-throughput targeted quantitation of U/HPLC data	https://github.com/Tarskin/HappyTools	[66]
GlycoDigest	Exoglycosidase prediction tool	http://www.glycodigest.org	[63]
Capillary Electrophoresis			
GUcal	Automatically calculates the glucose unit values for glycans in an electropherogram with a concomitant database search for interpretation and structural assignment.	www.gucal.hu	[73]
glyXalign	Designed for multiplex CE with the ability to align electropherograms based on an internal standard.	www.glyxera.com	[67]
GlycanAssure	Commercial Applied Biosystems: Integrated CE data acquisition and analysis software	https://www.thermofisher.com/order/catalog/product/A30841	
32 Karat	Commercial Sciex: 32 Karat Software with Fast Glycan Plugin: The software integrates the peaks, and then identifies the glycans using an internal database. fast-glycan-kit-application-guide.pdf.	https://sciex.com/Documents/tech%20notes/	

GlycoProfileAssigner [60], extends autoGU [61,62], to monitor peak movements after step-wise treatment of exoglycosidases that are ordered from least to most digested, and an approximate GU structure-based assignment step denotes an initial list of glycan structures that is refined by using the exoglycosidase digestion rules packaged in GlycoDigest [41,63]. GlycanAnalyzer [64•] improves upon GlycoProfileAssigner by using a novel scoring and regression model to rank the most likely peak movements that result from enzyme digestions [65]. An assigned glycan score depends on a number of factors such as optimised regression models that predict its expected GU shift, whether all masses are found in each profile of the exoglycosidase array, and the GU similarity to average values in GlycoStore. HappyTools is a standalone tool for performing all the steps of a completely automated data analysis workflow, including peak detection, retention time calibration and peak quantitation [66]. As reported by the authors of GlycanAnalyser scoring algorithms identify patterns of peak movement and assign structures based on the GU match either after treatment with a single or panel of exoglycosidases — typically a score is generated based on the difference between observed and known GU shifts. GlycanAnalyzer uses GlycoStore to calculate a GU similarity score and a regression model approach for peak shift prediction to generate an overall GU Score; the model is not fluorescent label/tag specific. Confidence in these assignments can be improved by complementing LC peaks shifts with mass data that is changes in reported glycan masses after digestion. For a detailed explanation readers are referred to [64•].

Capillary electrophoresis

Because of the outstanding separation power of CE for structurally related molecules, it is commonly used for the analysis of released glycans [67]. Currently, three different methods have been developed for CE-based structural elucidation of glycans: 1) calibration of migration time with molecular ladders and standards for database search, 2) exoglycosidase based sequential digestion, and 3) coupling CE with mass spectrometry. While CE instrumentation and sample preparation workflows are well established the required bioinformatics, tools are lagging.

Database-based identification of glycan structures requires accurate, robust and repeatable migration time determination; however, even the most accurate CE separations suffer from slight variability of migration times [68]. The GU approach, inspired by Kovats retention index in gas chromatography [69], overcomes this issue by comparing the migration times of unknown analytes with a mixture of oligosaccharide structures (dubbed as 'ladder') to assist structural elucidation [70,71].

While the GU approach for CE was described at the end of the last millennium, its automated utilisation is still limited. This approach has been improved by the Guttman group by introducing the so-called triple-internal standard method (TSM), which alleviates the need for a malto-oligosaccharide ladder [71]. Considering the migration times of co-injected standards of maltose, maltotriose, and maltopentadecaose – which are bracketing the glycan peaks of interest – an accurate virtual ladder can be set up and used in the subsequent data processing steps.

This method has been implemented in GUcal application [72,73] (together with the traditional internal / bracketing standard approach), which is a useful tool for structural elucidation of glycans with or without exoglycosidase digestions and is linked with GlycoStore [74], but it lacks the ability to process multiple runs at the same time in an automated fashion. In comparison the glyXera supports multilane CE migration time alignment for capillary gel electrophoresis-based *N*-glycan analysis [67] unfortunately, it does not address GU value calculation or structural elucidation.

Mass spectrometry

In recent years advances in MS instrumentation and methods for sample preparation have resulted in a significant increase of MS-based approaches adopted within the glycomics community [75,76]. There are two main strategies for elucidating glycosylation information using MS techniques: (1) global characterisation after the release of glycans from glycoproteins and (2) characterization of glycopeptides after proteolytic digestion. Glycoprofiling of released glycans is particularly useful when a rapid analysis is required and when combined with knowledge of biological pathways, MS profiles can facilitate the elucidation of structural classes of glycans. To gain glycosylation site-specific information for individual glycoforms, the second method, glycopeptide analysis, which requires digestion of the protein using a protease is necessary. This method is generally advantageous because it provides information about both glycan composition and the site of the glycan's attachment. However, the structural variation of glycosylation at any one site usually cannot be determined by high-resolution MS data alone. It requires a combination of MS-based mapping and MS/MS sequencing of glycopeptides to generate sufficient information.

Glycopeptide analysis is a fast-growing field and many (open access and commercial) software applications have emerged that offer novel and promising capabilities. For a detailed review of applications available readers are referred to [37,77,78•] and Table 2. Each of these tools differs by computational strategies and many have been designed to support specific workflows and/or limited datasets. As described by Liang *et al.*, all share the same fundamentals: (i) accurate mass measurement in precursor MS spectra; (ii) deducing putative glycan composition and peptide backbone sequence from LC–MS/MS spectra; (iii) pattern matching the marker fragment ions for peptide backbone and glycan against theoretical spectra from protein and glycan database, or attempt *de novo* sequencing of the glycan moiety; and (iv) scoring [79].

Recent applications, published between 2018–2019, include DecoyDeveloper [80], GlycopeptideGraphMS [81], GlySeeker [82] and GRITS Toolbox [83]. Briefly, DecoyDeveloper is a web-based application that generates high volumes of decoy glycopeptides with low mass differences. It uses an internal glycan composition database to produce biologically relevant glycan and peptide sequences. GlycopeptideGraphMS uses a graph approach to convert LC–MS features into nodes based on glycan mass and retention time and selects nodes whose differences match a predefined list of monosaccharide or disaccharide residues (represented as graph edges). In this way, retention clustered nodes deferring by a glycan mass are connected to each other and shown to be *N*-glycopeptides with the same peptide backbone but different glycan compositions. GlySeeker is a standalone database search application that enables automated large-scale *N*-glycan identification from LC–MS/MS (HCD) spectra. GlySeeker can be used to visualise *N*-glycan structures annotations and provides a corresponding FDR score for the assignment. GRITS toolbox is a platform-independent application for storing and processing glycomics MS data along with associated sample preparation and experimental metadata. It uses a curated glycan database and the Glycomics Elucidation and Annotation Tool (GELATO) [84] module to interpret MS data using functions provided by GlycoWorkBench plus additional features including: different fragmentation settings for each MS level; prediction of ions resulting from neutral loss; and options to create adducts/ions formed by neutral exchange.

Ion mobility

Over the last decade, ion-mobility spectrometry (IM–MS) has been increasingly used for the separation and analysis of isomeric glycans and their conjugates [85,86]. Most techniques separate ions based on their mobility through an inert drift gas under the influence of a weak electric field. This can be converted to a rotationally averaged collision cross-section (CCS) — an intrinsic property of a given ion under the defined experimental conditions used. For a detailed description of different IM–MS instrumentation and techniques, readers are referred to [87]. As discussed by Gray *et al.*, these CCS values may differ based on the conformation of the glycan analyte enabling separation of isomers, therefore, providing isomeric species and/or structural information [28]. Importantly, CCS values can be compared to database values or theoretical derived values [88] providing an insight into potential gas-phase conformations. To this end two databases have been released in the last few years i) GlycoMob (released glycans and standards) [89] and ii) a database of glycopeptides, glycans, and peptides constructed by the Costello and Zaia groups [90].

Machine-learning and glycosylation site prediction tools

The experimental characterisation of glycosylation sites is technically demanding, and computational methods trained on glycoproteomics site-specific knowledge to accurately predict unknown/putative glycosylation sites would be invaluable. Over the last few years, advances in

Table 2

Software available for analysing and annotating glycan and glycopeptide data

Name	Description	Link	Reference
Byonic™	Commercial. Supports the detection of glycosylation sites by using a hybrid integrated database and de novo sequencing approach.	http://www.proteinmetrics.com/products/byonic/	[91]
Cartoonist	Labels MS peaks with cartoons for *N*-linked oligosaccharides released from glycoproteins. Uses isotope envelopes as a filter for mass matching.	Available upon request	[92]
DecoyDeveloper	A glycopeptide decoy generator.	http://glycopro.chem.ku.edu/J1.php	[80]
GlycanAnalysis	Plugin for Mass++ to annotate MS/MS spectra with structures registered in GlycomeDB and KEGG GLYCAN.	http://www.shimadzu.co.jp/aboutus/ms_r/soft.html	[93]
Glycoforest 1.0.	*de novo* algorithm for sequencing glycan structures based on MS/MS spectra	https://glycoforest.expasy.org	[94]
GlycoMaster DB	For *N*-linked glycopeptides. Searches a protein sequence database and a glycan structure database to determine best matching peptide and glycan combination.	http://www-novo.cs.uwaterloo.ca:8080/GlycoMasterDB	[95]
GlycoMod	Given an experimentally determined mass (glycan and/or glycopeptide sequence GlycoMod determines all possible glycan compositions.	web.expasy.org/glycomod/	[96]
GlycoPAT	Comprehensive glycopeptide analysis platform includes FDR and an Ensemble-type scoring.	https://virtualglycome.org/glycopat	[97]
GlycoPep DB	*N*-glycan compositional assignment of glycopeptides from MSn data. Uses a 'Smart searching' approach to assign glycan masses from in-house databases.	http://hexose.chem.ku.edu/sugar.php	[98]
GlycoPep Detector	Determines *N*-linked glycopeptide composition from ETD data.	http://glycopro.chem.ku.edu/ZZKHome.php	[99]
GlycoPep Grader	Determines *N*-linked glycopeptide composition. Requires user to define protein sequence and a set of theoretical glycans. Decoy glycopeptide library used for generating FDRs.	http://glycopro.chem.ku.edu/GPGHome.php	[100]
GlycoPeptide Finder	Lists potential amino acids and glycan compositions based on the mass of a putative glycopeptide, and whether it is *N*- or *O*-glycosylated. Uses deconvoluted and deisotoped CID/CAD/HCD spectra and filters for oxonium ions. Support for calculating FDR via a decoy strategy.	Available upon request	
GlycopeptideGraphMS	Theoretical graph tool to identify and annotate glycopeptide data from MS/MS spectra.	https://bitbucket.org/glycoaddict/glycopeptidegraphms	[81]
GlycoPeptideSearch	Identify *N*-glycan glycopeptide from CID spectra by matching peptide-glycan pairs from user defined peptide sequences, and glycan structures stored in GlycomeDB.	https://edwardslab.bmcb.georgetown.edu/trac/GlycoPeptideSearch/	[101]
GlycoProteomeAnalyzer (I-GPA)	*N*-glycoproteome software includes a database of peptides produced by *in silico* trypsin digestion, and FDR scoring based on a decoy method. Latest version provides web-access tool.	https://www.igpa.kr	[102]
GlycoWorkBench	Comprehensive resource for interpreting and annotating mass spectrometry data by matching experimental data against theoretical MS/MS fragmentation.	http://www.glycoworkbench.org/	[103]
GlycReSoft	Supervised and unsupervised prediction based on quantitative features derived from calculations generated by Decon2LS/DeconTools for individually processed LC/MS data files.	https://code.google.com/archive/p/glycresoft/	[104]
GlySeeker	*N*-glycan database search engine for large-scale identification and visualisation including interpretation of tandem mass spectra with an FDR control.	Available upon request	[82]
GPQuest	Spectral library matching algorithm for *N*-glycopeptides using HCD tandem MS spectra. Uses a library of deglycosylated peptides to classify and identify intact glycopeptide tandem mass spectra.	Not available	[105]

Table 2 (*Continued*)

Name	Description	Link	Reference
GRITS	Software package for storing and processing glycomics MS data along with associated experimental metadata.	http://www.grits-toolbox.org	[83]
LaCyTools	Automated data processing tool for high-throughput analysis of LC–MS glycoproteomics data.	https://github.com/Tarskin/LaCyTools	[106]
Mass spectrometry-based Automated Glycopeptide Identification platform (MAGIC)	Identifies intact glycopeptides from CID spectra. Performs a peptide database search and uses a mass shift look-up table (compiled from biosynthetic rules) to score glycan compositions.	http://ms.iis.sinica.edu.tw/COmics/Software_MAGIC.html	[107]
MultiGlycan	Tool can be used to analysis MALDI and ESI spectra. It matches experimental isotopic envelopes with theoretical envelopes of *N*-glycans.	http://darwin.informatics.indiana.edu/MultiGlycan/	[108]
Peptoonist	Similar to Cartoonist but optimised for glycopeptides.	Not available	[109]
pGlyco	Intact *N*-glycopeptides analysis, provides access to a dedicated search engine and conducts comprehensive quality control including false discovery rate evaluation at the glycan, peptide and glycopeptide level.	http://pfind.ict.ac.cn/download/pGlyco/	[110]
SimGlycan	Commercial. Annotates experimental MS data by matching against an internal database and generates a scored list of candidate structures.	http://www.premierbiosoft.com/glycan/	[111]
Sweet-Heart	Supervised machine learning to identify *N*-glycopeptides from ion trap based LC–MS/MS.		[79]
UNIFI	Commercial. Uses an in-house library and GlycoBase to match glycans based on m/z for MS and Glucose Units for LC data sets.	www.waters.com	

Table 3

Site-prediction tools

Software	Description	Content	Predictor type	Link	Reference
GlycoEP	*N*-glycosylated, *O*-glycosylated and *C*-glycosylated	Sequence and predicted structural information of eukaryotic glycoproteins using a Support Vector Machine classifier	Sequence-based	http://crdd.osdd.net/raghava/glycoep/	[112]
GlycoMine	*N*-glycosylated, *C*-glycosylated and *O*- glycosylated	Random Forest classifier for site-specific prediction trained with human glycoprotein sequences.	Sequence-based	http://www.structbioinfor.org/Lab/GlycoMine/	[113]
GlycoMine ^{struct}	*N*-glycosylated and *O*- glycosylated	Random Forest classifier using a database of human 3D structures.	Structure-based	http://glycomine.erc.monash.edu/Lab/GlycoMine_Struct/	[114]
GlycoPP	*N*-glycosylated and *O*- glycosylated	Designed for prokaryotic site prediction using a Support Vector Machine classifier.	Sequence-based	http://crdd.osdd.net/raghava/glycopp/	[115]
GPP	*N*-glycosylated and *O*- glycosylated	Random Forest classifier using data from *O*-GlycBase 6.0	Sequence-based	http://comp.chem.nottingham.ac.uk/glyco/	[116]
NetNGlyc	*N*-glycosylated	Uses Artificial Neural Networks to predict human *N*-glycosylation sites.	Sequence-based	http://www.cbs.dtu.dk/services/NetNGlyc/	
NetOGlyc	*O*-glycosylated	Neural network trained with data from 12 human cell lines consisting of 600 proteins with approx. 3000 *O*-glycosites	Sequence-based	http://www.cbs.dtu.dk/services/NetOGlyc/	[117]
NGlycPred	*N*-glycosylated	Structure and residue pattern information from 154 eukaryotic structures uses a Random Forest classifier	Structure-based	https://bioinformatics.niaid.nih.gov/nglycpred/	[118]
SPRINT-Gly	*N*-glycosylated and *O*- glycosylated	Deep Neural Network and Support Vector Machines for predicting *N*-linked and *O*-linked glycosylated sites, respectively. Training data obtained from multiple databases.	Sequence-based	http://sparks-lab.org/server/SPRINT-Gly/	[119]

Table 4

Databases relevant for interpreting glycomics and glycoproteomics data

Database Name	Description	Link	Reference
Carbohydrate Structure Database (CSDB)	Carbohydrate structure database merged from bacterial, archaeal, plant and fungal databases. Provides access to bibliographic, taxonomic, NMR spectroscopic information.	http://www.csdb.glycoscience.ru	[127]
CSDB_GT	A curated database of glycosyltransferases found in *A. thaliana* and *E. coli*	http://csdb.glycoscience.ru/gt.html	[128•]
dbPTM	A database of protein post-translational modifications including glycosylation.	http://dbptm.mbc.nctu.edu.tw/	[129]
ECODAB (Escherichia coli *O*-antigen database)	A database focused on *O*-polysaccharide structures of *E. coli* lipopolysaccharides, with supporting 1H and 13C NMR chemical shift data.	http://www.nevyn.organ.su.se/ECODAB	[130]
EK3D	A curated database of K antigens from various *E. coli* serotypes plus 3D structures/model entries.	http://www.iith.ac.in/EK3D/	[131]
Glyco3D	A collection of databases that provide information on the three-dimensional features of oligosaccharides, glycosyltransferases, lectins, antibodies, and glycosaminoglycan-binding proteins.	http://glyco3d.cermav.cnrs.fr	[132]
GlycoDomainViewer	Web application that displays *O*-/*N*-linked glycosylation sites on proteins based on UniProt annotations and in-house data collections.	http://glycodomain.glycomics.ku.dk	[133••]
GlycoMob	Ion mobility collision cross section values for released glycans standards and their fragments.	http://www.glycomob.org	[89]
GlycoNAVI	Chemical synthesis of carbohydrate molecules, NMR spectra, activity and GBP interactions extracted from the literature.	https://glyconavi.org/Glycans/	
GlyConnect	A platform integrating sources of information to help characterise the molecular components of protein glycosylation	https://glyconnect.expasy.org	[134]
GlycoProtDB	Database of mouse *N*-glycoproteins from curated experimental data using IGOT approach.	https://acgg.asia/gpdb2/	[135]
GLYCOSCIENCE.DE	Provides access to a number of databases and bioinformatics tools that support structure, mass spectrometry, NMR and PDB querying, searching and analysis.	http://www.glycosciences.de	[136]
GlyCosmos	Provides access to glycosciences-related data including glycan-related genes, proteins, glycoconjugates, diseases, and pathway. Also serves to integrate a collection of databases developed by JCCGB.	https://glycosmos.org	
GlycoStore	Database focused on UPLC, HPLC, CE and RP-HPLC elution positions for over 600 glycan structures with links to taxonomy, glycoprotein and supporting literature.	https://www.glycostore.org	[42]
GlyGen	GlyGen is an international data integration and dissemination project for carbohydrate and glycoconjugate related data.	https://www.glygen.org	
GlyTouCan	Glycan sequence repository that denotes a unique accession number to each entry.	http://www.glytoucan.org	[137]
Kyoto Encyclopedia of Genes and Genomes (KEGG)	A database of genome sequences and other high-throughput data including glycan biosynthesis and metabolic pathways, roles of glycans in cellular processes, relation to disease, carbohydrate-binding proteins, and a structure database (KEGG GLYCAN).	http://www.genome.jp/kegg	[138]
MatrixDB	Database focused on interactions between extracellular matrix proteins and glycosaminoglycans.	http://matrixdb.univ-lyon1.fr	[139]
MonosaccharideDB	A comprehensive database of monosaccharides descriptions.	http://www.monosaccharidedb.org	
O-glycoproteome DB	Database of *O*-glycoproteins/peptides including information on the site of glycosylation.	http://www.glycoproteomics.someee.com/	
SugarBind	A curated database providing access to lectin adhesins of viral and bacterial pathogens and biotoxins and known carbohydrate ligands.	http://sugarbind.expasy.org	[140]
The Carbohydrate-Active enZYmes Database (CAZy)	Contains genomic, structural and biochemical information on glyco-enzymes.	http://www.cazy.org	[141]
The Consortium for Functional Glycomics (CFG)	Provides access to MALDI-MS glycan spectra from mouse and human tissue, and an extensive library of glycan array and gene microarray screening data sets.	http://www.functionalglycomics.org	[142]

Table 4 (*Continued*)

Database Name	Description	Link	Reference
UniCarb-DB	An LC–MS/MS glycan fragmentation database.	http://www.unicarb-db.org	[143]
UniCarbKB	A curated database of information on the glycan structures of glycoproteins reported in scientific literature.	http://www.unicarbkb.org	[144]
UniLectin3D	Classified list of lectins, their 3D structures, species information, and glycan affinity.	https://www.unilectin.eu	[145]
Unipep	Database describing theoretical *N*-linked glycosylation sites derived from UniProt that have been mapped to LC–MS data.	http://www.unipep.org	[146]
Glycan array and modelling software			
GLAD	GLycan Array Dashboard is a web-based tool to visualise, analyse, present, and mine glycan microarray data.	https://glycotoolkit.com/Tools/GLAD/	[147]
Gly-Spec	A web-based tool for predicting glycan specificity by integrating glycan array screening data and 3D structure.	www.glycam.org	[148]
GLYCAM-Web	A platform dedicated to simplifying the modelling and visualisation of carbohydrates and glycoconjugates.	www.glycam.org	[149–151]
Glycan Library (Glycosciences Laboratory -Imperial College London)	Library of glycolipid and lipid derivatised glycans (*N*-/*O*-linked glycans, lewis antigens, GAGs polysaccharides, etc) probes for microarrays.	http://www.imperial.ac.uk/glycosciences/	
GlycoPattern	A web-based tool to support the analysis of glycan array data for the Consortium for Functional Glycomics	https://glycopattern.emory.edu/	[152]
Privateer	Software for the conformational validation of carbohydrate structures	http://www.ccp4.ac.uk/html/privateer.html	[153]
Rosetta	A tool for correcting conformational and configurational errors in carbohydrates	https://www.rosettacommons.org	[154]

machine-learning algorithms have improved the accuracy of glycosylation site prediction (Table 3). There are two types of prediction methods: i) sequence-based predictors systematically assess the importance of predicted structural features as well as sequence information using protein sequence as the only input and ii) structure-based predictors that systematically assess the importance of structural features, for example, local secondary structure, solvent accessibility in addition to sequence information. These tools include GlycoMine, NetNGlyc, NetOGlyc, EnsembleGly, GPP, and SPRINT-Gly whose algorithms are based on sequence-derived features, for example, the physical/chemical properties of amino acids, position-specific scoring matrices, and predicted structural information. Only a few tools, namely, NGlycPred and GlycoMine^{struct} systematically assess the importance of structural features in glycosylation site prediction. Fewer structure-based predictors are available due to the limited number of experimentally solved structures. In general, the accuracy of sequence-based approaches outperforms structure-based and a detailed comparison was performed by SPRINT-Gly. In brief, for human *N*-linked prediction, it was reported that most tools have an accuracy performance above 0.9, but Matthews correlation coefficient (MCC) varied considerably between 0.53–0.94. The performance of the latest method SPRINT-Gly is even higher than these structure-based techniques, highlighting the importance of using a larger training dataset and novel predicted structural features. By providing a site prediction probability, these prediction tools complement the identification of novel sites of glycosylation identified by mass spectrometry approaches.

International glycoinformatics infrastructure

In 2008, the NIH workgroup 'Frontiers in glycomics' highlight the need for curated and sustainably funded glyco-structure databases. It was expected that any structural or glycoconjugate database(s) would provide metadata and supporting experimental data (Table 4). To this end, four major resources have grown to provide long-term solutions for centralising glyco-related content including GlyGen (http://www.glygen.org/) [120], Glycomics@ExPASy (https://www.expasy.org/glycomics), GlyCosmos (https://glycosmos.org/), and the Carbohydrate Structure Database (http://csdb.glycoscience.ru).

International efforts to improve data interoperability across the glycosciences have led to the development and release of frameworks for reading, converting and manipulating glycan structures, for example, glypy [121•], REStLESS (ResiduEs as Smiles and LinkagEs as SmartS) [122], EUROCarbDB GlycoCT and IUPAC supported converters [123,124], and GlycanFormatConverter [125]. This is an important prerequisite due to the diversity of encoding formats available (GlycoCT, WURCS, IUPAC, SMILES, and InChi). For a comprehensive description of structure encoding formats, readers are referred to [126].

Reporting guidelines

In 2011 the MIRAGE (Minimum Information Required for A Glycomics Experiment) was launched to provide guidelines for the reporting of glycomics data: sample preparation [155], MS [156], glycan microarray [157], and liquid chromatography analysis [158•]. These guidelines provide a

framework for the efficient documentation of data collections and subsequently glycoinformaitc activities to promote data sharing and reproducibility by using ontologies and standardised formats, which are now being integrated into databases, for example, GlycoStore (https://glycostore.org/mirage) and UniCarb-DR (https://unicarb-dr.biomedicine.gu.se/generate).

Discussion

Currently, our understanding of the structures, synthesis and biochemical functions of glycans and glyconjugates is limited. Nevertheless, advances in glycoanalytics are producing large volumes of data that needs to be organised and interpreted. In this short review, we have described advances and availability of resources for analysing glycomics and glycoproteomics data collections. In the last few years, several international activities have started to improve data interoperability between glycosciences and other omic disciplines, for example, GlyGen is bringing together information from multiple carbohydrate and glycoconjugate related data sources. It is these efforts that will help address bioinformatics questions posed by researchers and drive advances in the field.

Conflict of interest statement

Nothing declared.

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